# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A two-digit number exceeds the number formed by reversing its digits by 36. The difference of its digits is: 1

**(a)** 3 **(b)** 5 **(c)** 36 **(d)** 4

**2.** The perimeter of a rectangle is 108 cm and the length is 5 times the breadth. Its area is: 1

**(a)** 414 cm² **(b)** 405 cm² **(c)** 387 cm² **(d)** 729 cm²

**3.** A polyhedron has 30 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

**(a)** 12 **(b)** 32 **(c)** 10 **(d)** 20

**4.** The mean of 25 values was found to be 24. Later it was found that 90 was misread as 40. The correct mean is: 1

**(a)** 74 **(b)** 22 **(c)** 24 **(d)** 26

**5.** If $x + \frac{1}{x} = 4$, then $x^2 + \frac{1}{x^2}$ equals: 1

**(a)** 8 **(b)** 18 **(c)** 14 **(d)** 16

**6.** An isosceles trapezium has parallel sides 12 cm and 22 cm and non-parallel sides 13 cm. Its area is: 1

**(a)** 408 cm² **(b)** 221 cm² **(c)** 204 cm² **(d)** 214 cm²

**7.** If every corner of a cube is sliced off, the number of faces of the resulting solid is: 1

**(a)** 12 **(b)** 6 **(c)** 8 **(d)** 14

**8.** A class has 24 students with mean age 12 years. When the teacher's age is included, the mean rises to 13. The teacher's age (years) is: 1

**(a)** 37 **(b)** 13 **(c)** 36 **(d)** 32

**9.** The sum of the dates in a $2 \times 2$ calendar block is 64. The smallest date in the block is: 1

**(a)** 11 **(b)** 20 **(c)** 13 **(d)** 12

**10.** The length of a rectangle is increased by 25% and its breadth decreased by 20%. If the original area is 2000 cm², the new area (in cm²) is: 1

**(a)** 2100 **(b)** 2500 **(c)** 2000 **(d)** 2200

**11.** A cube is painted all over and cut into unit cubes. If exactly 54 of the unit cubes have exactly one face painted, the side of the big cube is: 1

**(a)** 4 **(b)** 6 **(c)** 5 **(d)** 7

**12.** The mean of the first 15 odd natural numbers is: 1

**(a)** 8 **(b)** 16 **(c)** 15 **(d)** 225

**13.** **Assertion (A):** The expression $3x + 5$ has the value 17 when $x = 4$. 1

**Reason (R):** Terms with the same variable part are called like terms.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**14.** **Assertion (A):** The area of a triangle with base 10 cm and height 8 cm is 40 $cm^2$. 1

**Reason (R):** The area of a rectangle of length 10 cm and breadth 8 cm is 80 $cm^2$.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**15.** **Assertion (A):** A hexagonal prism has 18 edges. 1

**Reason (R):** A hexagonal prism has 6 vertices.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Prove that $(n+3)^2 - (n-3)^2 = 12n$ and use it to find the value of $(5+3)^2 - (5-3)^2$. 2

**17.** The area of a triangle is 135 cm² and one side is 18 cm. Find the altitude on that side and the area of a triangle with the same base and double the altitude. 2

**18.** A cube is painted all over and cut into unit cubes. If 84 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

**19.** The mean of 6 numbers is 35. When one number is removed, the mean of the remaining 5 numbers is 32. Find the removed number. 2

**20.** The sum of the digits of a two-digit number is 11 and the tens digit exceeds the ones digit by 5. Find the number. 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** The sum of a two-digit number and its reverse is 132 and the digits differ by 2. Find the number (tens digit larger). 3

**22.** The perimeter of a rectangular plot is 180 m and its length is 4 times its breadth. Find its area and the cost of fencing at ₹ 12 per metre. 3

**23.** A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 168? Hence find the number with exactly one painted face. 3

**24.** The mean age of 15 players is 20 years. When the coach is included the mean becomes 22. Find the age of the coach. 3

**OR**

If $x + \frac{1}{x} = 4$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

**25.** An isosceles trapezium has parallel sides 20 cm and 44 cm and each slant side 20 cm. Find its height and area. 3

**OR**

In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 1000. How many shaded triangles are there at that stage? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26. Case study.** Aarav performs a number trick: *Think of a number $n$. Multiply it by 6. Add 12. Divide by 3. Subtract twice your original number. The answer is always 4.* 5

(i) Write the expression after multiplying by 6.

(ii) Write it after adding 12.

(iii) Write it after dividing by 3.

(iv) Complete the algebra to show the answer is always 4 and check with $n = 7$.

**27.** Prove that $(3n + 2)^2 - (3n - 2)^2 = 24n$ and use it to evaluate it for $n = 5$. 5

**OR**

The area of a rhombus is 384 cm² and one diagonal is 24 cm. Find the other diagonal, the side, and the altitude. 5

## END OF QUESTION PAPER — Total: 50 Marks

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